(Please write your Enrolment No. immediately) Enrolment No. _____

# MID TERM EXAMINATION

## B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

**Third Semester, November 2022**

**Paper Code: CIC-209** **Subject: Data Structures**

**Time: 1½ Hrs.** **Max. Marks: 30**

**Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

| Q. No. | Question | Max. Marks | CO(s) |
|---|---|---|---|
| 1 | **Attempt the following:** | | |
| (a) | Briefly explain Abstract Data Type model with the help of a diagram. | 2 | CO1 |
| (b) | Why are data structures required? List various types of data structures. | 2 | CO1 |
| (c) | What is sparse matrix? Explain briefly the 3-tuple (or triplet) representation of sparse matrix with an example. | 2 | CO2 |
| (d) | Differentiate between static implementation (using arrays) and dynamic implementation (using pointers) of a list data structure. | 2 | CO2 |
| (e) | Define the terms depth, height, internal node, and leaf node with respect to binary trees with the help of an example. | 2 | CO2 |
| | | | |
| 2 (a) | Write algorithms for:<br>(i) insertion of an element in a sorted singly linked list, and<br>(ii) deletion of an element from a sorted singly linked list. | 5 | CO2 |
| (b) | Consider the following Preorder and Inorder Traversals of a binary tree:<br>Preorder : A B D E C F<br>Inorder : D B E A F C<br>You are required to construct the binary tree based on above given traversals. Show all the steps to construct the tree. | 5 | CO2 |
| | | | |
| 3 (a) | What is binary search tree? Write the algorithms for Preorder, Inorder and Postorder Traversals of the tree. | 5 | CO2 |
| (b) | Why circular queue preferred over simple queue? Demonstrate and explain with the help of a suitable example. | 5 | CO2 |
| | | | |
| 4 (a) | Convert the following infix expression into postfix expression using stack:<br>A + B * C – D / (E * F) ^ G | 5 | CO2 |
| (b) | Construct the binary search tree using the following data elements:<br>35, 70, 50, 10, 48, 80, 55, 11, 85<br>Show binary search tree after insertion of each data element. | 5 | CO2 |